

1 Introduction

Over the last decade tremendous progress has been made in the field of artificial intelligence (AI). From self-driving cars to protein folding: machine learning (ML) has shown itself to be inevitable for future technology. But not only has it transformed the industry world, it has also become an integral part of scientific research. The processing and analysis of huge amounts of data, which are typical for experiments in high energy or astrophysics, pose major challenges to traditional methods. With the rise of AI, we can tackle these tasks in a reasonable way. But what about theoretical physics? The questions there might not involve a lot of data, but could be even more difficult to answer. They are however of great importance for the future, because they can give us a deep understanding of nature, which is very often the basis for the invention of new technologies.

One such topic, which has attracted a lot of interest, is the holographic principle. It appeared first in the context of string theory and states, that quantum gravity can be encoded on the boundary of its spacetime. It can therefore be thought of as a hologram of the information stored in lower dimensional space. It turns out, that this actually defines a non-gravitational theory on the boundary, which should be equivalent to the higher-dimensional bulk theory. Two physical theories without any apparent connection are somehow ultimately the same. This is also called the gauge-gravity duality, because the boundary system is often found to be a gauge theory. The gravity theory is related to it in a special way: whenever one theory is in the strong coupling regime, the other is weakly coupled. This opens up the possibility to analyze a system at energy scales, at which perturbation theory would usually break down. To do that we simply have to find and use its holographic dual. If this relationship actually holds, then we have a powerful tool at hand to access information about gravitational systems or even quantum gravity itself. Black holes are examples for such systems, which can be explored via holography. As a matter of fact, the holographic principle was inspired from a result about the entropy of black holes. It was found that the maximal information contained in a black hole scales with its boundary surface [4]. On a more practical side, the duality could enable us to investigate strong coupling phenomena of the gauge theories like holographic

QCD. Although this theory is not the same as the physical QCD with $SU(3)$ color gauge group, it could still give us insights about aspects common to QCDs. An important concept is for example the confinement of quarks, which is still puzzling to this day. An understanding of its underlying mechanism is tightly intertwined to the question of stability of protons. With the holographic principle we might be able to tackle these difficult tasks. Unfortunately, it is not clear how to find the precise mapping between the two descriptions of the same physics.

There exist various versions and formulations of the principle and the most popular duality is arguably the AdS/CFT correspondence. Its general statement is that any conformal field theory (CFT) defined on a conformally flat spacetime of d dimensions is equivalent to a quantum gravity theory in a bulk spacetime, which is asymptotically $AdS_{d+1} \times C$, with C being some compact manifold. Although this correspondence has been analyzed in great detail over the last two decades, it still remains unproven. The equivalence between gauge theory and gravity is only conjectured. Could machine learning perhaps help us to gain a deeper understanding of the correspondence?

Since the appearance of neural networks many architectures have been introduced, which have produced impressive results. Of particular interest are the generative models, which can learn a (sometimes even unknown) probability distribution and create samples from it. Such networks are for example variational autoencoders (VAEs), Boltzmann machines, generative adversarial networks (GANs) or normalizing flows. The ability to sample efficiently from a learned distribution is especially important, when the original probability density cannot be computed due to its complexity. This is for example the case for quantum field theories, where the partition function can only be calculated for Gaussian distributions. In this situation flow networks can help us. By using this network architecture to approximate the probability distribution, we can not only sample, but also can the learned density be directly accessed, i.e. it is tractable. This is one of the reasons, why normalizing flows will be our network of choice.

In order to apply this network to holography, we have to make use of a peculiar fact about the interior of the bulk space. As mentioned above, the CFT is defined on the AdS boundary. When we move from the boundary into the bulk space along the radial (or holographic) direction, which is perpendicular to the boundary surface, we can find another CFT. This new theory turns out to be a renormalized version of the one on the boundary. The flow into the bulk interior corresponds to an RG-flow of the CFT and the radial coordinate can be identified with the inverse RG-scale. If we can set up a neural network, which can perform an RG-flow on a field theory, then we might find an

emergent geometric structure in the network corresponding to a hyperbolic space.

In the following we will first study the AdS/CFT correspondence in more detail to gain an better understanding of the physical framework. After that, the flow networks are reviewed alongside with features to be used later. We are then ready to build an RG-flow neural network and apply it to a discrete QFT on a lattice. The training of various networks is discussed and methods to extract geometrical information from the flow are presented. At last we will mention some applications for the trained flow and show possible extensions of the model.

2 AdS/CFT Correspondence

A first concrete example of the holographic principle was given by Maldacena in 1997 [9], when he conjectured the equivalence between a $\mathcal{N} = 4$ supersymmetric $SU(N)$ Yang-Mills theory with conformal symmetry and string theory on $AdS_5 \times S^5$. This was motivated by the fact, that by taking specific limits of type IIB string theory in a background curved by N coincident D3-branes one can get either one of the two theories (see [6] for a detailed review). The equivalence is already apparent at the level of symmetry. Since the two theories are conjectured to be the same, their symmetry groups have to coincide. In general, the conformal field theory lives in a d dimensional flat spacetime and the gravity theory is defined in a $d + 1$ dimensional hyperbolic background. A hyperbolic space is usually defined as a hypersurface of a higher dimensional flat space. This means that an AdS_5 space inherits the Lorentz $SO(4, 2)$ symmetry from its embedding, which has 15 generators. The symmetries of a CFT on the other hand are generated by the usual 10 Lorentz generators corresponding to translations, rotations and boosts with additional 5 conformal symmetries, which include four special conformal and one scale transformation. These 15 generators satisfy the same algebra as the isometry group of the AdS space. In fact, $SO(4, 2)$ is also called the conformal group. On the gravity side we further have a $SO(6)$ symmetry from the compactification on the sphere S^5 . This corresponds to $\mathcal{N} = 4$ supersymmetry of the field theory, which has symmetry group $SU(4)$, because the two groups are isomorphic to each other: $SO(6) \cong SU(4)$.

To see that the CFT actually lives on the boundary of the bulk space, we look at AdS_{d+1} -metric global coordinates [3]

$$ds^2 = \frac{L^2}{\cos^2 \rho} (-d\tau^2 + d\rho^2 + \sin^2 \rho d\Omega^2) , \quad (2.1)$$

with $-\pi < \tau \leq \pi$, $0 \leq \rho < \frac{\pi}{2}$ and $\sum_{i=1}^d \Omega_i^2 = 1$. We observe that the boundary is the hypersurface at $\rho = \frac{\pi}{2}$ and has the topology of $\mathbb{R} \times S^{d-1}$, which is conformal to d -dimensional Minkowski space. Therefore, a conformally invariant QFT can be thought of as living on the conformal boundary of AdS_{d+1} . It contains the same amount of infor-

mation as the gravity theory. Or in other words, the bulk theory can be holographically projected and encoded onto the boundary surface. With a change of coordinates and a conformal rescaling we can bring the metric into the simple form

$$ds^2 = \frac{L^2}{z^2} (dz^2 + \eta_{ij} dx^i dx^j) , \quad (2.2)$$

which is referred to as Poincaré coordinates and covers half of AdS_{d+1} . This is most common and convenient metric of AdS space. Its boundary lies at $z = 0$ with Minkowski metric η_{ij} .

For easier calculation and implementation we will choose to use the Euclidean version of AdS space, the hyperbolic space $\mathbb{H}^{d+1}$. We will however refer to this space as AdS, since a Euclidean metric can be easily obtained from the Minkowski version by Wick rotation to imaginary time. With this, the boundary metric becomes $\eta_{ij} = \delta_{ij}$. Bulk indices will be denoted by greek letters μ, ν and boundary indices by latin letters i, j . We will use $x, y \in \mathbb{R}^d$ as coordinates of the boundary with $x = (x^0, \dots, x^d)$ and $z \in \mathbb{R}$ will usually be the holographic radial direction.

To analyze the correspondence we would like to find some observables, which we can compute in both theories. Comparing results can then give us a first check, whether the holographic principle is realized at all. A simple quantity to calculate is the two-point function or correlator, which will be our focus.

2.1 Conformal two-point function

Due to the large group of Poincaré and conformal transformations, the general form of the one, two and three-point functions are completely determined by symmetry considerations up to constant factors. Suppose we have two complex fields $\psi_1(x), \psi_2(y)$ with conformal dimensions Δ_1, Δ_2 on the boundary. Because of translation invariance, we already know that the correlator between the two fields has to be a function of the distance

$$\langle \psi_1(x) \psi_2^*(y) \rangle = f(|x - y|) . \quad (2.3)$$

In a CFT we are allowed to perform scaling transformations $x \rightarrow \lambda x$ without changing the observables. A field will transform under scaling like

$$\psi_i(\lambda x) = \lambda^{\Delta_i} \psi_i(x) . \quad (2.4)$$

We can then restrict the form of the two-point function. The constraint reads

$$\begin{aligned} \langle \lambda^{\Delta_1} \psi_1(\lambda x) \lambda^{\Delta_2} \psi_2^*(\lambda y) \rangle &= \lambda^{\Delta_1 + \Delta_2} f(\lambda(x - y)) \\ &\stackrel{!}{=} f(x - y) \end{aligned} \quad (2.5)$$

which is satisfied, when f decays according to a power-law in the distance $|x - y|$:

$$f(|x - y|) \sim \frac{1}{|x - y|^{\Delta_1 + \Delta_2}} \quad (2.6)$$

One can further find a relation between the conformal dimensions Δ_1 and Δ_2 . A symmetry, that plays a very important role in CFTs, is the invariance under inversions

$$x \rightarrow \frac{x}{|x|^2} . \quad (2.7)$$

After performing this transformation on the correlator and imposing invariance, we find the condition:

$$\begin{aligned} \left\langle \frac{1}{x^{2\Delta_1} y^{2\Delta_2}} \psi_1\left(-\frac{1}{x}\right) \psi_2^*\left(-\frac{1}{y}\right) \right\rangle &= \frac{1}{x^{2\Delta_1} y^{2\Delta_2}} \frac{c}{\left(-\frac{1}{x} + \frac{1}{y}\right)^{\Delta_1 + \Delta_2}} \\ &\stackrel{!}{=} \frac{c}{|x - y|^{\Delta_1 + \Delta_2}} \end{aligned} \quad (2.8)$$

To fulfill this constraint, we have to set $\Delta_1 = \Delta_2$. This means, that the correlator between two fields is only non-zero, if their dimensions are the same. So in general, the two-point function in a conformal field theory will always be an algebraic decay in the form

$$\langle \psi(x) \psi^*(y) \rangle \sim \frac{1}{|x - y|^{2\Delta}} \quad (2.9)$$

This gives us a practical way to check, if a theory is not or cannot be described by a CFT. When a physical system is in a highly disordered state, the correlation function will usually be an exponential decay with some finite correlation length. A power-law implies then, that the correlation length has (formally) diverged.

2.2 Boundary values as sources

A first precise formulation of the correspondence was given by Witten [11], where he showed how to calculate the two-point function of a CFT using the bulk gravity action. Assume that for operators $\mathcal{O}$ in the CFT there exist a term in the field theory action,

that couples them to an external source J . In general, the interaction between them has the form

$$\int dx \mathcal{O}(x) J(x) \quad (2.10)$$

The usual procedure for a QFT is to construct a partition function $Z[J]$ and calculate the n-point functions as functional derivatives of $Z[J]$ with respect to the external source at $J = 0$. In the following, we will express the bulk action in terms of the boundary values of the classical bulk fields.

As the bulk theory, consider the case of a free complex scalar field $\phi(x, z)$ in a general background with action

$$S_{\text{bulk}}[\phi, g_{\mu\nu}] = \int d^d x dz \sqrt{g} [g^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi + m^2 |\phi|^2 + R] . \quad (2.11)$$

To find the classical equation of motion for ϕ , we integrate by parts. After fixing the geometry to be AdS_{d+1} , we get for the ϕ -dependent terms

$$S_{\text{bulk}}[\phi] = - \int d^d x dz \sqrt{g} \phi^* (-\square_g + m^2) \phi + \int d^d x dz \partial_\mu (\sqrt{g} g^{\mu\nu} \phi^* \partial_\nu \phi) \quad (2.12)$$

with the Laplacian

$$\begin{aligned} \square_g &= \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu) \\ &= \frac{z^2}{L^2} (\partial_z^2 + (1-d) \frac{1}{z} \partial_z + \partial_x^2) . \end{aligned} \quad (2.13)$$

By imposing $\delta S = 0$, we find the equation of motion from the first term in equation (2.12). The second term has no contribution, because it can be written as a surface integral by invoking Stokes' theorem. The e.o.m. then reads

$$z^2 \partial_z^2 \phi - (d-1) z \partial_z \phi + (z^2 \partial_x^2 - L^2 m^2) \phi = 0 , \quad (2.14)$$

where the general solution will be a convolution of a boundary configuration and the Green's function K satisfying

$$(-\square_g + m^2) K(x, z) = \delta(x, z) . \quad (2.15)$$

To solve this, we look for a solution of the homogeneous equation, which has a singularity at one point on the boundary. This corresponds to the behavior we would expect for a δ -function. Since we have translation invariance in x direction, the Green's function

should only be a function of z and we take as an ansatz

$$K(z) = z^\alpha . \quad (2.16)$$

By changing coordinates using the inversion map $x \rightarrow \frac{x}{z^2 + |x|^2}$, $K(z)$ becomes

$$K(z) = \frac{z^\alpha}{(z^2 + |x|^2)^\alpha} . \quad (2.17)$$

This function has a singularity at $x = 0$ for $z \rightarrow 0$, which means it behaves like a δ -function on the boundary. We can now express our solution as the convolution of the Green's function with a source:

$$\phi(x, z) \sim \int dy K(x - y, z) \phi_0(y) , \quad (2.18)$$

where $\phi_0(x) \equiv \phi(x, z = 0)$.

We can use this expression to write the action as a functional of the boundary values of $\phi(x, z)$. As mentioned above, the second term in the action (2.12) can be expressed as a surface integral with induced metric h_{ij} and normal vector n^μ as

$$\int d^d x \sqrt{h} \phi^* n^\mu \partial_\mu \phi = \int d^d x \frac{1}{z^{d-1}} \phi^* \partial_z \phi . \quad (2.19)$$

The derivative with respect to z can be estimated to lowest order in z from the classical solution (2.18) as

$$\partial_z \phi \sim z^{d-1} \int dy \frac{\phi_0(y)}{|x - y|^{2\alpha}} . \quad (2.20)$$

With this, the action on the boundary will be of the form

$$S_{\text{bulk}}[\phi_0] = \int dx dy \frac{\phi_0^*(x) \phi_0(y)}{|x - y|^{2\alpha}} . \quad (2.21)$$

From this expression, the two-point function for the CFT can be calculated.

Assume, that we have the following coupling of a complex boundary field $\psi(x)$ with an external source $\phi_0(x)$:

$$- \int dx [\psi(x) \phi_0^*(x) + \psi^*(x) \phi_0(x)] . \quad (2.22)$$

Since the AdS/CFT correspondence states, that the boundary and bulk theories are the

equivalent, their partition functions should contain the same information, i.e.

$$Z_{\text{CFT}} = Z_{\text{bulk}} \approx e^{-S_{\text{bulk}}[\phi_0]} . \quad (2.23)$$

In the saddle point approximation of the gravity theory, the classical bulk action plays the role of the generating functional for the field theory and we can compute the two-point function for ψ by taking a functional derivatives of Z_{bulk} w.r.t. the sources ϕ_0 :

$$\begin{aligned} \langle \psi^*(x) \psi(y) \rangle &= \frac{\delta}{\delta \phi_0(x)} \frac{\delta}{\delta \phi_0^*(y)} Z_{\text{bulk}}[\phi_0] \\ &\sim \frac{1}{|x - y|^{2\alpha}} . \end{aligned} \quad (2.24)$$

Thus, we have found a polynomial decay of the correlation function, which is consistent with the result from the CFT, and can identify the exponent with the conformal dimension, $\alpha = \Delta$. We have obtained way to calculate any n-point function of the QFT without precise knowledge of its action or dynamics, since the information is provided by the bulk theory. One should note, that this is the weakest form of the AdS/CFT correspondence, because it is only valid in the saddle point approximation of the bulk theory, e.g. the supergravity limit of string theory. In the case of the original correspondence involving SUSY Yang-Mills theory with gauge group $SU(N)$, this implies the large N limit $N \rightarrow \infty$. However, the equivalence is believed to hold even in the quantum regime and on all energy scales.

2.3 Holographic renormalization

When calculating observables of a QFT, one usually has to do them carefully in a renormalized fashion, as quantities can easily diverge due to the presence of arbitrary small scales or infinite volumes in the definition of the bare action. A potential divergence is for example the integration $\int d^d x$ over the boundary at $z = 0$, since the area is infinite. Before we perform any calculations, we have to regularize the theory by introducing a cutoff to the length or energy scale, e.g. area integral goes over a surface at some $z_0 \neq 0$. We can send $z \rightarrow 0$ after the calculation to remove the cutoff. This transformation is part of the isometry $(x, z) \rightarrow (\lambda x, \lambda z)$ of AdS space, which is dual to the dilation symmetry of CFTs. For the boundary theory this corresponds to a coarse-graining, which induces an RG-flow.

As an example let's consider again a massive field $\phi(x, z)$ in the bulk. A typical correlation

function between two points in the bulk at $z = z_0$ will decay exponentially with length of paths connecting them [7] as

$$\langle \phi^*(x) \phi(y) \rangle_{z=z_0} \sim \sum_{\text{paths}(x,y)} e^{-m\mathcal{D}(x,y)} . \quad (2.25)$$

This expression will be dominated by the geodesic between the points (x, z_0) and (y, z_0) for $m \gg L^{-1}$, such that

$$\langle \phi^*(x) \phi(y) \rangle_{z=z_0} \sim e^{-m\mathcal{D}_{\text{geo}}(x,y)} . \quad (2.26)$$

The geodesic in AdS will be of logarithmic dependence on $|x - y|$ and is given by

$$\mathcal{D}_{\text{geo}}(x, y) = L \ln \left(\frac{\left(|x - y| + \sqrt{|x - y|^2 + z_0^2} \right)^2}{z_0^2} \right) . \quad (2.27)$$

For large distances $|x - y| \gg z_0$, this can be approximated as

$$\mathcal{D}_{\text{geo}}(x, y) \approx L \ln |x - y| . \quad (2.28)$$

We now see, that the two-point function in a hyperbolic space will be approximately a power-law given by

$$\langle \phi^*(x) \phi(y) \rangle_{z=z_0} \sim \frac{1}{|x - y|^{2mL}} . \quad (2.29)$$

This is the familiar form, that we get for a CFT with identification $\Delta = mL$. However, this functional dependence is only approximate and the precise behavior at short scales is modified and is not polynomial. This implies, that the long distance physics of the bulk theory is related to the UV of the boundary. This is called the IR-UV relation or duality, which was first discussed in [10]. The theory, that we find at z_0 is, due to the effective UV-cutoff induced by the IR-cutoff in the bulk, the renormalized CFT at energy scale

$$\Lambda_0 = \frac{1}{z_0} . \quad (2.30)$$

This implies a relation of the renormalization group flow of the CFT with the radial flow of the gravity theory, see figure 2.1. The information of the interior space enclosed by the dotted line is encoded on its surface area.

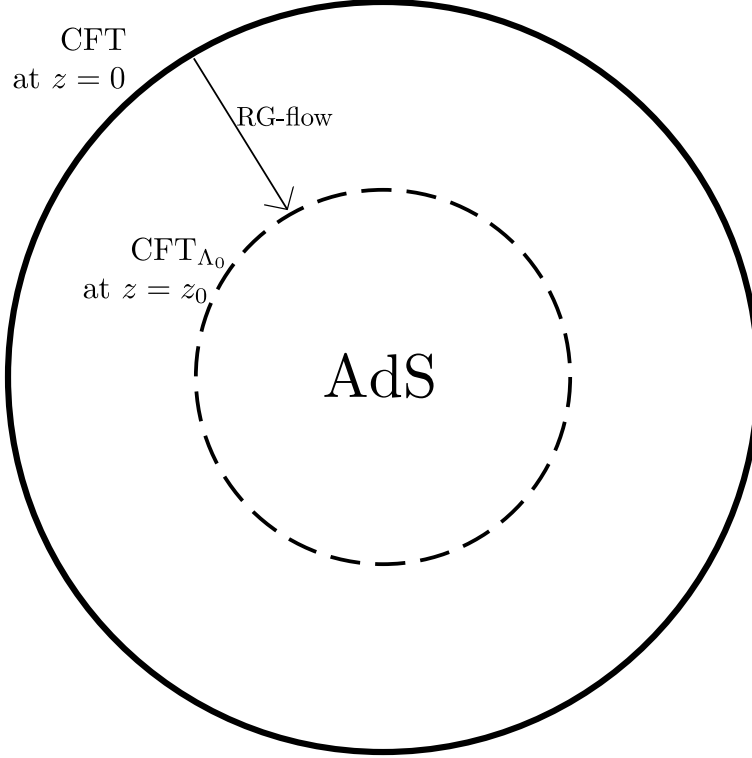


Figure 2.1: Moving from in radial direction z into the interior of AdS space is equivalent to renormalizing the boundary CFT

In fact, one can make this relationship more concrete. But as with the calculation of the two-point function, a precise computation is only possible in the saddle point approximation.

In a Wilsonian approach we would integrate out the UV degrees of freedom in the CFT, which should correspond to an integration of the long wavelength modes in the bulk [8]. In the classical limit, we can then define an inner correspondence of the CFT at z_0 and the interior volume of AdS space by integrating out bulk fields in the excluded area for $z < z_0$ [2].

A more precise mapping can be achieved when we go into the Hamilton-Jacobi formulation of the gravity theory, where the first order flow equations can be cast into the form of Callan-Symanzik equation [5]. In a semiclassical treatment the field values $\phi(x, z)$ can be interpreted as the running coupling constants of the CFT. The Hamilton equation of motions of the bulk fields are then equivalent to the renormalization group flow equations of the CFT [1]:

$$z \frac{d}{dz} \phi = \frac{\delta \mathcal{H}}{\delta \pi} \leftrightarrow \Lambda \frac{d}{d\Lambda} g = \beta(g) , \quad (2.31)$$

where $\mathcal{H}$ is the Hamiltonian with conjugate momentum π and g is some running coupling

constant in the CFT. The metric of the bulk is therefore a metric of the space of RG-flow. However, such a mapping is only valid in the semiclassical limit and also depends on the renormalization scheme. That means that we have to take precautions when setting up a coarse-graining procedure for a CFT, if we want the RG-flow space to be described a metric. Because there exist various renormalization schemes, not all RG-flow spaces will exhibit a geometric structure even if there exist a dual gravity description for the CFT.

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